B E N C H M A R K R E P O R T

Five Models Shut the Door Before Checking What's Behind It

When LLMs Choose Action Over Assessment in a Night Shift Emergency

Behavioral Analysis of 22 LLMs on an Emergency Response Sequencing Question

22	77.3%	3	5/5
LLMs Tested	Pass Rate	Chose Panic Shutoff	Put E Wrong

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1. Executive Summary

I posed an emergency response sequencing question to 22 LLMs: you are a night shift operator alone at a WWTP at 2 AM. Severe sludge bulking is causing active washout over the clarifier weirs. Put five response actions in the correct order. The correct sequence follows a fundamental protocol: assess the situation first (B, C, D in any order), notify the team with complete information (A), then take drastic action with supervisor approval (E — shutting the plant inlet). Six valid sequences exist.

This is the first sequencing question in the series. All previous twelve benchmarks required selecting a single correct option. Here, models must order five steps according to operational protocol — a fundamentally different cognitive task that tests procedural reasoning rather than factual knowledge.

The pass rate is 77.3% (17/22). This places the benchmark in the mid-range tier, alongside six other benchmarks in the 63–77% cluster. After thirteen benchmarks, the cumulative funnel remains at zero: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0 → 0 → 0 → 0 → 0 → 0.

F1 — All Five Failures Put E in the Wrong Position: Every failing model placed the inlet shutoff (E) somewhere other than last. Three models put E first ("panic shutoff"); two put E fourth instead of fifth ("one step slip"). No model failed on the assessment ordering or notification timing. The entire benchmark's discriminating power lies in a single decision: where to place E.

F2 — Three Models Chose "Panic Shutoff": Gemini 2.5 Pro, Gemini 3 Pro Preview, and DeepSeek R1 placed E first. All three framed the situation as an active environmental violation requiring immediate containment. Their reasoning was internally coherent — "stop the pollution first" — but it violates the operational principle that shutting a plant inlet at 2 AM without assessment or approval has cascading consequences: sewer overflows, pump station alarms, and upstream flooding.

F3 — R1's Second Genuine Wrong Answer: DeepSeek R1 produced an 8,010-character think block and deliberately concluded "E first, then assess." This is not a parsing artifact — R1 genuinely chose the wrong sequence. After eleven benchmarks of "correct reasoning, lost to formatting," R1 has now produced two consecutive genuine wrong answers (#11 and #13). The model's narrative has shifted from formatting victim to substantive fallibility.

F4 — Gemma Achieves 100% for the First Time: All four Gemma models passed, including Gemma 1B and 4B — both of which failed on the previous benchmark. This is Gemma's first clean sweep in the series. The family that has been the most variable now demonstrates that procedural sequencing may be within its capability even when material selection and design criteria are not.

F5 — Two Failure Clusters, One Root Cause: The five failures form two distinct clusters: "panic shutoff" (E first — 3 models) and "one step slip" (E fourth instead of fifth — 2 models).

Both clusters share a common root: underestimating the consequences of shutting a plant inlet without authorization. The difference is degree — the panic cluster skips assessment entirely, while the slip cluster completes assessment but acts before notifying.

2. The Question

The scenario places the model in the role of a solitary night shift operator facing an active emergency. The five actions represent three categories: assessment (B, C, D), communication (A), and drastic intervention (E). The correct protocol follows a clear hierarchy: gather facts, report with data, act with approval.

The trap is the urgency of the scenario. “Active sludge washout” and “significant solids escaping to effluent” create pressure to act immediately. Models that prioritize stopping the pollution over understanding the situation place E too early. The prompt states that the manager and engineer are “reachable via phone/WhatsApp” — a signal that communication is expected before drastic action.

With 5 items to sequence, there are 120 possible orderings. Six are correct (assessment in any order, then A, then E). Random correct probability is 5%.

Table 1: The Five Response Actions

Step	Category	Description
B	Assessment	Check the SCADA system — real-time plant data, DO levels, flow rates, MLSS
C	Assessment	Check inlet sensors and readings — determine if abnormal influent is the cause
D	Assessment	Physically inspect inlet water — color, odor, foam (field verification)
A	Communication	Notify the team via WhatsApp — report findings, request guidance
E	Intervention	Shut off the plant inlet — drastic action, requires supervisor approval

Table 1: The three assessment steps (B, C, D) can appear in any order. Communication (A) must follow all assessment. Intervention (E) must be last. This yields 6 valid sequences out of 120 possible orderings.

A note on question format: this is the first sequencing question in the WWTP series. All previous twelve benchmarks asked models to select a single correct option. Here, models must arrange five steps in order — testing procedural reasoning rather than factual recall. The cognitive demand is different: instead of evaluating options against criteria, models must construct a temporal chain where each step’s position depends on the others.

3. Results

Figure 1: Sequence Distribution

Sequence	Count	Status
B, C, D, A, E	7	(valid)

B, D, C, A, E	7	**(valid)**
D, C, B, A, E	2	**(valid)**
C, D, B, A, E	1	**(valid)**
B, C, D, E, A	2	**(E before A)**
E, B, D, C, A	2	**(panic shutoff)**
E, D, C, B, A	1	**(panic shutoff)**

Figure 1: Four valid sequences account for 17 models. Three invalid sequences account for 5 models. The two dominant valid sequences (B,C,D,A,E and B,D,C,A,E) differ only in whether SCADA or physical inspection comes second.

Table 2: Complete Results — All 22 Models

Model	Sequence	Status	Out Tok	Latency	Family
Claude Haiku 4.5	B,C,D,A,E	**PASS**	366	5.4s	Claude
Claude Opus 4.1	B,C,D,E,A	**FAIL**	343	15.2s	Claude
Claude Opus 4.5	B,D,C,A,E	**PASS**	434	12.6s	Claude
Claude Sonnet 4	B,D,C,A,E	**PASS**	325	10.3s	Claude
Claude Sonnet 4.5	B,D,C,A,E	**PASS**	372	12.4s	Claude
Gemini 2.0 Flash	C,D,B,A,E	**PASS**	506	5.0s	Gemini
Gemini 2.0 Flash Lite	B,C,D,A,E	**PASS**	369	4.1s	Gemini
Gemini 2.5 Flash	B,C,D,A,E	**PASS**	525	22.2s	Gemini
Gemini 2.5 Pro	E,B,D,C,A	**FAIL**	555	26.1s	Gemini
Gemini 3 Flash Preview	B,C,D,E,A	**FAIL**	353	14.6s	Gemini
Gemini 3 Pro Preview	E,D,C,B,A	**FAIL**	483	95.1s	Gemini
Gemma 3 12B	B,D,C,A,E	**PASS**	~370	12.6s	Gemma
Gemma 3 1B	B,D,C,A,E	**PASS**	288	6.8s	Gemma
Gemma 3 27B	B,D,C,A,E	**PASS**	~350	8.7s	Gemma
Gemma 3 4B	B,C,D,A,E	**PASS**	~350	7.8s	Gemma
DeepSeek R1 (CoT)	E,B,D,C,A	**FAIL**	2,020	12.7s	DeepSeek
DeepSeek V3.1	D,C,B,A,E	**PASS**	498	4.3s	DeepSeek
DeepSeek V3.2	D,C,B,A,E	**PASS**	295	7.8s	DeepSeek
Qwen3 235B A22B Instruct	B,C,D,A,E	**PASS**	397	4.4s	Qwen
Qwen3 Coder 480B	B,C,D,A,E	**PASS**	298	3.0s	Qwen
Qwen3 Next 80B Instruct	B,D,C,A,E	**PASS**	522	3.7s	Qwen
Qwen3 Next 80B Thinking	B,C,D,A,E	**PASS**	5,851	33.1s	Qwen

Table 2: All 22 models. Note that R1's failure is a genuine wrong answer, not a parsing artifact — the model deliberately chose E first in its reasoning.

4. LLM Behavior Analysis

4.1 Two Failure Clusters, One Root Cause

The five failures form two distinct clusters that share a single root cause: placing the inlet shutoff before it should occur.

Table 3: Failure Cluster Analysis

Cluster	E Position	Models	Behavior
"Panic Shutoff"	E first	Gemini 2.5 Pro, Gemini 3 Pro Preview, DeepSeek R1	Skips assessment entirely. Treats active washout as requiring immediate containment.
"One Step Slip"	E fourth (before A)	Claude Opus 4.1, Gemini 3 Flash Preview	Completes assessment correctly but shuts inlet before notifying supervisor.

Table 3: Both clusters share the same root cause: underestimating the operational consequences of shutting a plant inlet without proper authorization.

The panic shutoff cluster is the more dangerous of the two. Gemini 2.5 Pro framed the scenario as “an active environmental discharge” requiring “immediate containment.” Gemini 3 Pro Preview called it “an active environmental violation.” DeepSeek R1 concluded that “immediate action is needed to halt further hydraulic loading.” All three models prioritized stopping the perceived environmental harm over understanding what was causing it.

This reasoning sounds compelling — and in many emergency contexts, “stop the source” is the correct first instinct. But in wastewater operations, shutting the plant inlet at 2 AM is not a simple switch. It stops flow to the treatment plant while the collection system continues to receive wastewater. Within hours, pump stations overflow, manholes back up, and untreated sewage can enter streets or waterways. The “fix” creates a larger environmental problem than the one it addresses.

The one step slip cluster is a milder error. Claude Opus 4.1 and Gemini 3 Flash Preview both completed the assessment correctly (B, C, D) but placed E before A. They assessed the situation, gathered data — and then shut the inlet without calling the supervisor. The assessment was correct; the chain of command was violated. In operational terms, this is an unauthorized action: shutting a plant inlet is a decision that requires supervisor approval, not a field call an operator makes alone at 2 AM.

4.2 R1’s Narrative Shift — Second Consecutive Wrong Answer

DeepSeek R1 produced an 8,010-character think block that explicitly debated the sequencing. The model considered placing assessment first but concluded: “Immediate action is needed to halt further hydraulic loading and prevent additional solids from overwhelming the system.” It chose E, B, D, C, A — shutoff first, then assess, then notify last.

This is not a parsing artifact. R1's response after the think block begins with "E, B, D, C, A" in plain text. The model deliberately chose this sequence. Combined with the Odor Pipe benchmark (#11), where R1 genuinely selected FRP, this marks two consecutive benchmarks where R1 has been substantively wrong.

Across thirteen benchmarks, R1's track record now includes: formatting issues causing score loss (8 benchmarks), genuine wrong answers (2 benchmarks — #11 and #13), and correct answers (3 benchmarks — #1, #4 pending, and #12). The narrative that defined R1 for the first ten benchmarks — "excellent reasoning, lost to formatting" — is no longer the complete picture. R1 can reason incorrectly, and when it does, its extensive think block provides a detailed record of how the wrong conclusion was reached.

4.3 Gemini's Generational Split — Fourth Occurrence

Gemini produced the widest within-family split on this benchmark: 3 pass, 3 fail. The pattern is now familiar but with a twist.

Table 4: Gemini Model Results

Model	Sequence	Status	Note
Gemini 2.0 Flash	C,D,B,A,E	PASS	Old
Gemini 2.0 Flash Lite	B,C,D,A,E	PASS	Old
Gemini 2.5 Flash	B,C,D,A,E	PASS	New
Gemini 2.5 Pro	E,B,D,C,A	FAIL	New
Gemini 3 Flash Preview	B,C,D,E,A	FAIL	Newest
Gemini 3 Pro Preview	E,D,C,B,A	FAIL	Newest

Table 4: The two Pro models (2.5 Pro and 3 Pro Preview) chose panic shutoff. Both Flash variants (2.5 Flash and 3 Flash Preview) attempted the correct sequence but 3 Flash slipped E one position early.

This is the fourth benchmark with a Gemini generational split. A new pattern appears to be emerging: the Pro variants fail while the Flash variants pass or come closer to passing. Both Pro models chose "panic shutoff" (E first), framing the scenario as an environmental emergency requiring immediate action. The Flash models, even when wrong (3 Flash Preview), at least completed the assessment phase first.

Whether this reflects different training emphases between Pro and Flash variants, or whether Pro models are more susceptible to urgency framing, cannot be determined from this data. The observation is noted as a pattern to track.

4.4 Gemma's First 100% — After Consecutive Failures

All four Gemma models passed this benchmark, including Gemma 1B and 4B — both of which failed on Primary Clarifier (#12). Gemma 1B, which selected the dangerous opposite answer (F

— open top) on the previous benchmark, produced a correct and well-reasoned sequence here: B, D, C, A, E.

This is Gemma's first clean sweep in thirteen benchmarks. The family that has oscillated between 0% (#8) and 100% (#13), between dangerous answers (#6, #12) and correct reasoning, demonstrates once again that performance is question-dependent. Procedural sequencing appears to be a task type that small Gemma models can handle, even when they struggle with material selection or design criteria questions.

4.5 Family Performance Overview

Figure 2: Family Pass Rates

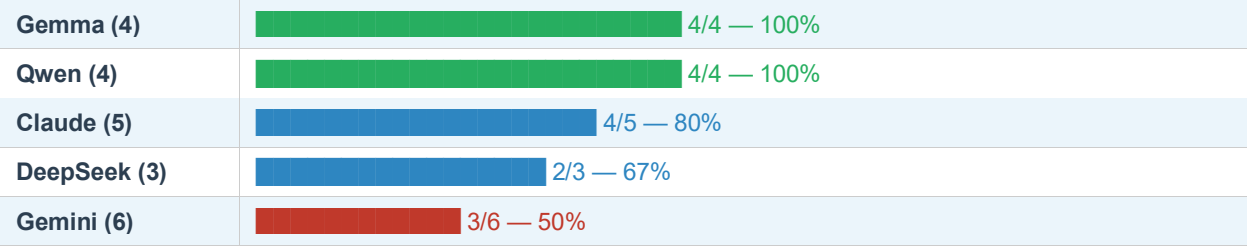

Figure 2: Gemma and Qwen achieve 100%. Gemini is the worst performer at 50% — dragged down by both Pro variants choosing panic shutoff.

The family rankings on this benchmark are notably different from the previous two. Gemma, the worst performer on #12 (50%), is now the joint-best. Gemini, which was 100% on #12, drops to the worst position at 50%. Claude, which was 100% on #12, drops to 80% on a single Opus 4.1 error. The role reversals continue: no family's position is stable across benchmarks.

4.6 Trap Taxonomy Update — Thirteen Benchmarks

Table 5: Updated Trap Taxonomy

Benchmark	Trap Name	Pass %	Failure Mode
#1 Material Selection	—	100%	—
#2 Pump RCA	Electrical Trap	77%	Electrical anchoring
#3 Dewatering RCA	Local Trap	68%	Local anchoring
#4 Biogas	Aerobic Trap	14%	Aerobic anchoring
#5 Walkway	Corrosion Trap (FRP)	36%	Chemical resist. anchoring
#6 Confined Space	"Almost Safe"	68%	Completeness illusion
#7 Sample Preserv.	Acid Trap	95.5%	Protocol anchoring
#8 Tool Selection	"False Safe"	68.2%	Hazard blindness
#9 Emergency Flocc.	Coagulant Trap	68.2%	Mechanism confusion
#10 SCADA Cabling	Cost-Reliability	63.6%	Partial knowledge

#11 Odor Pipe	UV Anchoring (FRP)	27.3%	Judgment error
#12 Primary Clarifier	Shape Trap	86.4%	Small model reasoning
#13 Sludge Bulking	Urgency Trap	77.3%	Action-over-assessment

Table 5: Thirteen benchmarks. The Urgency Trap is the first to test procedural sequencing rather than factual selection. The "action-over-assessment" failure mode is new to the taxonomy.

5. Cross-Benchmark Comparison — Thirteen Questions Deep

Figure 3: Pass Rate Across 13 WWTP Benchmarks

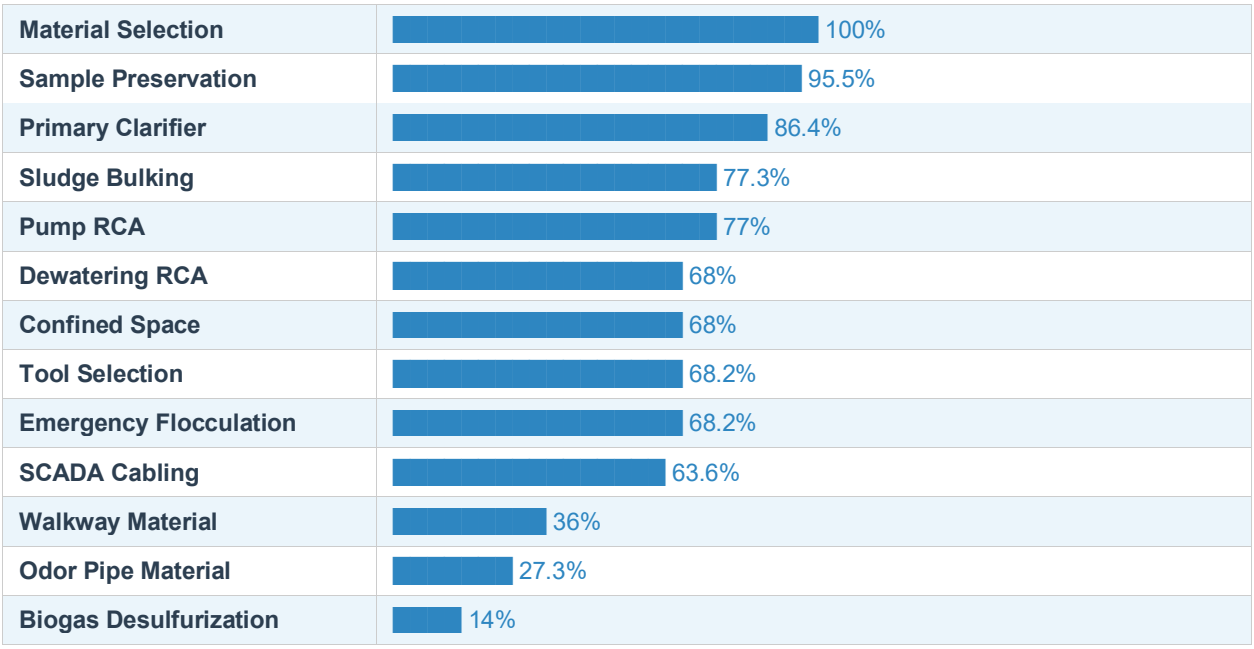

Figure 3: Thirteen benchmarks spanning 14% to 100%. Sludge Bulking enters the mid-range cluster at 77.3%.

The mid-range cluster now contains seven benchmarks (63–77%), making it the largest tier. The Sludge Bulking benchmark enters at the top of this cluster, near the boundary with the easy tier. Despite being a fundamentally different question format (sequencing vs. selection), its pass rate falls within the same range as the six single-choice mid-range benchmarks. The first sequencing question in the series produced a pass rate indistinguishable from the single-choice mid-range benchmarks, suggesting that procedural ordering and factual selection may present comparable difficulty levels for current LLMs — the cognitive task changed, but the performance band did not.

The cumulative funnel remains at zero for the ninth consecutive benchmark: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0 → 0 → 0 → 0 → 0 → 0. No model has passed all thirteen questions.

6. Conclusions

This benchmark introduces a new question format to the WWTP series — procedural sequencing rather than single-option selection — and discovers that the series' most consistent failure pattern applies here too: when a scenario creates urgency, some models prioritize action over assessment. The "panic shutoff" cluster (three models placing E first) mirrors the anchoring-on-salient-features pattern documented throughout the trap taxonomy, with urgency replacing chemical properties or cost as the anchoring stimulus.

The fact that all five failures share a single root cause — misplacing E — reveals that operational protocol understanding is the discriminating factor on this question. Every model understood that assessment should precede notification. No model placed A first. The entire variation occurs around E: should the operator shut the inlet before notifying (unauthorized action) or after (protocol-compliant action)? This binary decision separated 77% of models from the 23% that failed.

R1's second consecutive genuine wrong answer marks a significant shift in the model's series narrative. The think block's reasoning — "halt further hydraulic loading" — is the same environmental-urgency framing used by the two Gemini Pro models. R1 did not fail because of formatting; it failed because it adopted the wrong decision framework. After ten benchmarks of formatting issues, the last two have shown that R1's reasoning itself can be compromised by urgency or anchoring.

Gemma's first 100% — on a procedural question, after failing a factual question — adds another dimension to the series' growing body of evidence that model capabilities are task-type dependent. Small models that cannot identify the correct material for a pipe or the correct design for a clarifier can correctly sequence an emergency response protocol. Knowledge and reasoning appear to be distinct capabilities, at least at the 1B–4B parameter scale.

7. Methodology Notes

Non-determinism: Single run per model. Results are indicative, not absolute.

Question format: This is the first sequencing question in the series. Models must order 5 steps, yielding 120 possible permutations. Six are correct (5% random probability). The observed 77.3% pass rate is approximately 15 times above this random chance baseline. All previous benchmarks used single-option selection.

Parsing: The platform extracts the first 5-letter sequence from each response. R1's failure is a genuine wrong answer, not a parsing artifact — the model's post-think response begins with "E, B, D, C, A" in plain text.

Token counts: Some Gemma models reported 0 output tokens in API metrics despite visible responses. Counts marked with ~ are estimated.

API conditions: All models accessed via Kaggle Benchmarks platform API with default temperature settings.

Series consistency: All thirteen benchmarks use the same platform and default temperature settings. This benchmark uses a different evaluation method (sequence matching vs. first-character extraction) but the same platform infrastructure.